An update on the Southern Bantu Resource Grammar project

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Stellenbosch University Iyunivesithi Universiteit

CSIR Touching lives through innovation

Overview

Part 1:	The GF mission and introduction
	About languages and linguistics
Part 2:	GF - our guiding principles
	The years 2009 - present
	Status quo
Part 3:	Selected topics
	What next?

Part 1: Introduction

The GF mission: “To formalize the grammars of the world and make them available for computer applications”

Including the grammars of the Bantu languages of South(ern) Africa

Our efforts, insights and plans so far...

This talk: about a number of resource grammars

The next talk: about a number of applications

About languages and linguistics

Languages[1]
World languages: 7159
Niger-Congo: 1554
(Narrow) Bantu: 546 (440-680)

Speakers
350 million
30% of the population of Africa
5% of the population of the world

1. https://www.ethnologue.com/

The Bantu languages

- A high level of shared features and linguistic continuity
- Strong similarities in grammar and vocabulary, indicating a common origin and more recent (4000 - 3000 years ago) shared history
- **A unique agreement feature using noun prefixes and concords**
- The majority are severely under-resourced: lack of political will; money; people; connectivity; data; LT and time to change all this
- A nice and, in our view, important challenge and opportunity!

Some facts

- The 1990s, democracy and a new SA Constitution
- 11 Official languages, 9 are Bantu languages
- Multilinguality enshrined in the Constitution
- Standardisation: PanSALB (1995) (https://www.pansalb.org/) and the various National Language Bodies
- The language of instruction ONLY up to the 3rd school grade (10 year olds)
- Language technology for the SA Bantu languages became relevant.

Some numbers

Language	Group	L1 speakers	Wikipedia	GT	GF RG
Zulu (zul)	Nguni	14 613 202	11 712	√	√
Xhosa (xho)	Nguni	9 786 928	2 290	√	√
Afrikaans (afr)		6 365 488	126 072	√	√
N.Sotho (nso)	Sotho-Tsn	5 972 255	8 774	√	√
English (eng)		5 288 301	7 043 021	√	√
Tswana (tsn)	Sotho-Tsn	4 972 787	2 400	√	√
S.Sotho (sot)	Sotho-Tsn	4 678 964	1 482	√	√
Tsonga (tso)		2 784 279	949	√	
Swati (ssw)	Nguni	1 692 719	1 125	√	√
Venda (ven)		1 480 565	824	√	√
Ndebele (nbl)	Nguni	1 044 377	212	√	

Some linguistics

Orthography: Slightly different, but linguistic structure is similar

Morphology: Complex, agglutinative, structurally similar

Basic word order: SVO

Grammatical structure: two fundamental systems

1. Nominal classification
2. Concordial agreement

Nominal classification

Nominal classification, the **system of noun classes**

The (Zulu) noun has two main parts: noun **prefix** and noun **stem**

Noun prefix consists of two parts: the **preprefix** and the **basic prefix**

Every noun belongs to a so-called noun class (its **class gender**)

Class gender generates **grammatical agreement** by means of the class prefixes

Noun classes are numbered - Zulu and Northern Sotho have 18 and 17 noun classes, resp.

The noun classes

Class.No.	Class.Pref.	Example (zul)	Class.Pref.	Example (nso)
1-2	um(u)-; aba-	umuntu; abantu (person)	mo-; ba-	motho; batho (person)
1a-2a	u-; o-	ubaba; obaba (father)	ϕ-; bo-	tate; botate (father)
3-4	um(u)-; imi-	umuzi,imizi (village)	mo-; me-	motse; metse (village)
5-6	i(li)-; ama-	ifu; amafu (cloud)	le-; ma-	leru; maru (cloud)
7-8	isi-; izi-	isikole; izikole (school)	se-; di-	sekolo; dikolo (school)
9-10	iN-; iziN-	inja; izinja (dog)	ϕ-/n-/m-; di-	mpša; dimpša (dog)
11	u(lu)-	uthando (love)		
14	ubu-	ubuhlungu (pain)	bo-	bohloko (pain)
15	uku-	ukudla (food; to eat))	go	go ja (to eat)
16	pha-	phansi (below)	fa-	fase (below)
17	ku-	kude (far)	go-	godimo (above)
18	mu-	muva (behind)	mo-	morago (behind)

Concordial agreement

Concordial agreement, the **system of concords**

They formally mark the relationship/**agreement** between a **noun** and **other words** in a sentence

This gender agreement must be observed in all parts of the utterance which are linked to the noun

Verbs, pronouns, adjectives, relatives, possessives etc. (SC, OC, AC, RC, PC etc.)

Close resemblance between class prefixes and concords.

The verb

Morphological structure of the verb:

	Neg1	**SC**	**Neg2**	**Temp**	**Asp**	**OC**	**Refl**	**ROOT**	**VExts**	**Term**	**Rel/Imp**
zul		ba			nga	zi		theng	is	a	
nso		ba			ka			rek	iš	a	
zul		ba			nge	zi		theng	is	e	
nso		ba			ka se			rek	iš	e	

Moods (both zul and nso): Indicative, situative, consecutive, imperative, subjunctive
Basic tenses (zul): Present, past, remote past, future, remote future
Basic tenses (nso): Present, past, future

Examples

Abantu	abaningi	bangazithengisa	izimoto	zabo
NPref2-NStem	AC2-AdjStem	SC2-Pot-OC10-VStem	NPref10-NStem	PC10-AbsPN2
People	who are many	they may sell them	cars	of them

'Many people may sell their cars.'

Batho	ba bantši	ba ka rekiša	dikoloi	tša bona
NPref2-NStem	Dem2 NPref2-AdjStem	SC2 Pot VStem	NPref10-NStem	PC10 AbsPN2

===

Abantu	abaningi	bangezithengise	izimoto	zabo
NPref2-NStem	AC2-AdjStem	SC2-PotNeg-OC10-VStem	NPref10-NStem	PC10-AbsPN2
People	who are many	they may not sell them	cars	of them

'Many people may not sell their cars.'

Batho	ba bantši	ba ka se rekiše	dikoloi	tša bona
NPref2-NStem	Dem2 NPref2-AdjStem	SC2 PotNeg VStem	NPref10-NStem	PC10 AbsPN2

Part 2: GF - our guiding principles

1. Be as true as possible to the **linguistics** of the language
2. Find and work with the best **linguists** so that the linguistic integrity of the artefact is of the highest possible standard
3. For under-resourced languages it is vital to become part of existing vibrant multilingual projects/**communities** - not to try to reinvent the wheel
4. For under-resourced languages, plan for the **long haul**

GF: starting out

2001 - 2008

Finite-state computational morphology

ZuMorph with XhoMorph, SswMorph and NblMorph bootstrapped from it TsnMorph

2009

Finding GF

EACL 2009 in Athens
GFSS 2009 in Gothenburg and the Foods grammars

2010

Visit to Gothenburg

Bootstrap the afr RG from the Dutch.

2013

GFSS on the Fraueninsel, Bavaria

From TsnMorph to tsn MiniRG

2015

GFSS 2015 in Malta

Improvements to the afr RG

GF: Digging in - our 10 year plan

2018

A bumper year!

Laurette’s PhD done!
GFSS in Stellenbosch
I retire!

2020

Bantu RGs

GF mission
Acquiring first funding
zul RG (a mission!)
Lionel Posthumus
Collecting grammar texts

2022

Fist SA Bantu RG

The zul RG contributed
Tackling the nso RG

2024

More RGs

ssw RG bootstrapped
xho RG bootstrapping starts

2025

nso RG

Inspired by the zul RG, based on the SBAS


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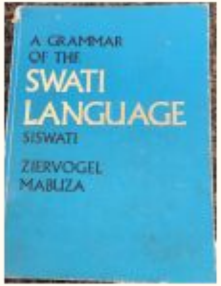
PC TALJAARD
JN KHUMALO
SE BOSCH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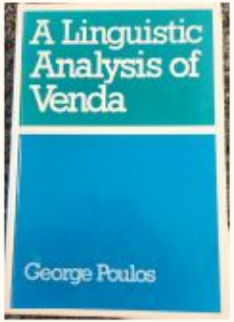
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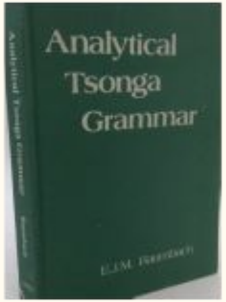
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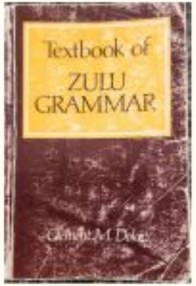
Textbook of
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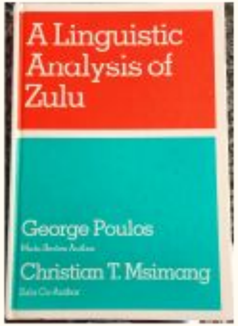
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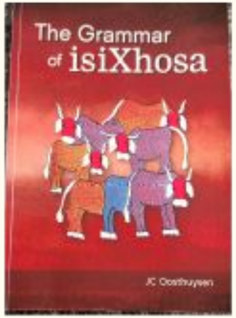
Christian T. Msimang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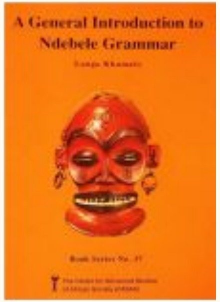
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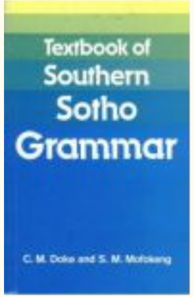
Textbook of
Southern
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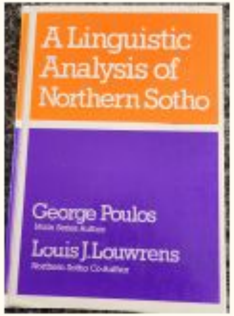
NOORD-SOTHO


A Linguistic Analysis of Northern Sotho
George Poulos
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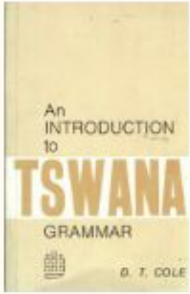
An
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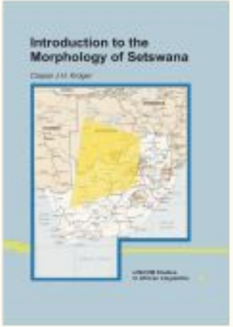
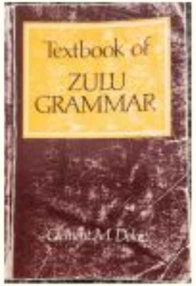
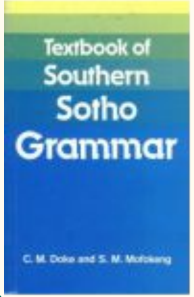
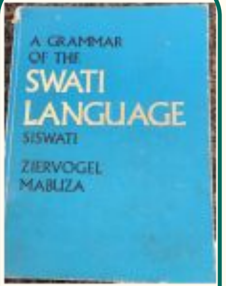
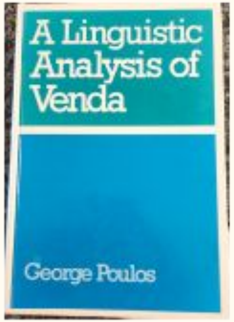
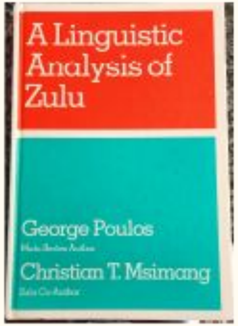

Introduction to the
Morphology of Setswana

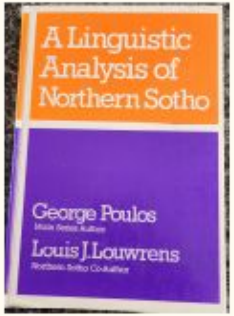




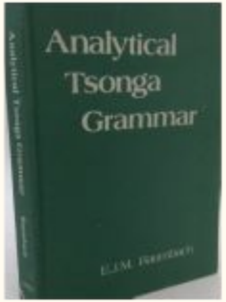


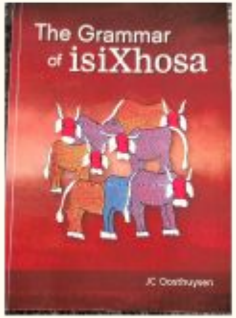


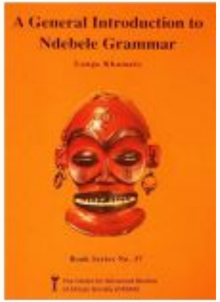


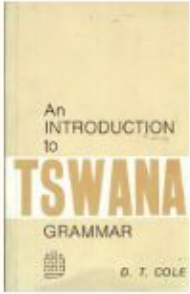


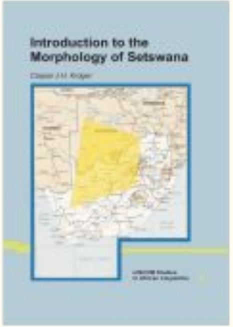
Linguists'
bootstrapping!









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Status quo: RGs

zul:

- Most mature RG
- Still lacks a few moods, the indirect verbal relative, auxiliary verbs, interrogatives.
- Various applications (Laurette's talk)
- Publications

ssw:

- Bootstrapped from the zul RG (systematic spelling changes[1])
- As mature as zul RG
- Not as yet application ready as the zul RG (lacks comprehensive lexicon)
- Publication

xho:

- Bootstrapped from the zul RG (systematic morphological and sound changes[1])

1. S E Bosch, L Pretorius and A Fleisch. Experimental bootstrapping of morphological analysers for Nguni languages. *Nordic Journal of African Studies*, 17(2):66-88, 2008.

Status quo (continued)

- **nso:**
- Almost as mature as the zul RG
- Uses the SBAS and inspired a few additions to it (e.g. the adverbials)
- Confirmation of the suitability of the SBAS.

tsn: To be bootstrapped from the nso RG in the near future.

ven: The next on our agenda of RGs. A next use of the SBAS.

nbl, sot: *Plan* to bootstrap

tso: *Plan* as “separate” effort, using the SBAS.

Part 3: Selected topics

- The categories and functions of the Southern Bantu RGs
- A treebank for testing and comparison
- Differences between zul and nso

Lexical Categories of the Southern Bantu RGs

Closed classes

Category	CAS?	Explanation
`A`	yes	adjective stem
`AdA`	yes	adjective-modifying adverb
`Adv`	yes	basic adverb
`Conj`	yes	conjunction
`ConjN`		conjunction between NPs
`LocN`		locative class noun stem
`Pron`	yes	absolute pronoun
`Quant`	yes	demonstrative
`QuantPron`		quantitative pronoun

Open classes

Category	CAS?	Explanation
`N`	yes	noun stem
`V`	yes	one-place verb stem
`V2`	yes	two-place verb stem
`V3`	yes	three-place verb stem
`V2V`	yes	verb stem with NP and V complement
`VV`	yes	verb stem with VP complement
`VS`	yes	verb stem with S complement

Phrasal Categories of the Southern Bantu RGs

Category	CAS?	Explanation
Utt	yes	utterance: sentence
Phr	yes	phrase in a text
Voc	yes	vocative
PConj	yes	phrase-beginning conjunction
S	yes	declarative sentence
Imp	yes	imperative
RS	yes	relative sentence
Temp	yes	tense (past, present or future)
Pol	yes	polarity (positive or negative)
Cl	yes	declarative clause
VP	yes	verb phrase
RCl	yes	relative clause
NP	yes	noun phrase
AP	yes	adjectival phrase
Loc		locative
LocAdv		locative adverb
CN	yes	common noun
Postdet		postdeterminer
Predet	yes	predeterminer
RP	yes	relativiser
Det	yes	fixes the number of a common noun
Num	yes	singular or plural

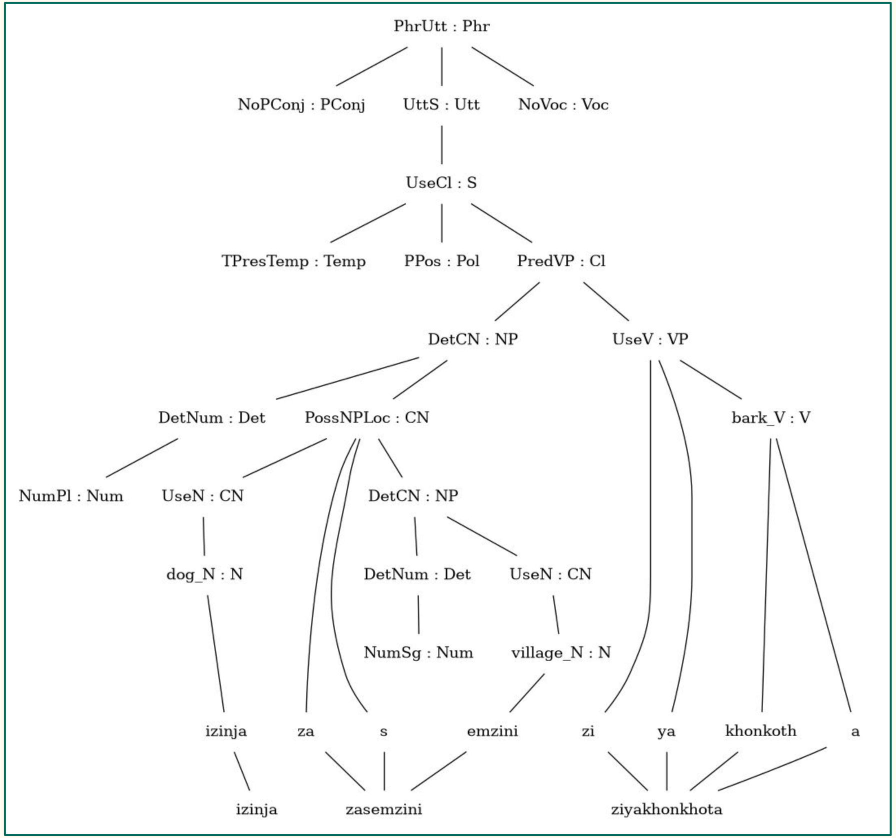
Functions of the Southern Bantu RGs

Value cat.	CAS	SBantu	Total
AP	2	-	2
Adv	1	5	6
LocAdv	-	9	9
CN	3	6	9
NP	4	4	8
Pron	-	2	2
Postdet	-	4	4
Predet	-	4	4
Det	-	1	1
LocN	-	1	1
Loc	-	1	1
VP	7	6	13
RP	1	3	4
Cl	1	-	1
RCl	1	-	1
S	2	2	4
RS	1	-	1
Imp	1	1	2
Utt	3	4	7
Phr	1	1	2
PConj	1	-	1
Voc	2	-	2
TOTAL	31	54	85

Treebank

Sentences from Taljaard and Bosch (Handbook of IsiZulu)

- Testing the bootstrapping process
- Illustrating similarities and differences (between Nguni languages and a Sotho groups)


PhrUtt : Phr
NoPConj : PConj
UttS : Utt
NoVoc : Voc
UseCl : S
TPresTemp : Temp
PPos : Pol
PredVP : Cl
DetCN : NP
UseV : VP
DetNum : Det
PossNPLoc : CN
bark_V : V
NumPl : Num
UseN : CN
DetCN : NP
dog_N : N
DetNum : Det
UseN : CN
NumSg : Num
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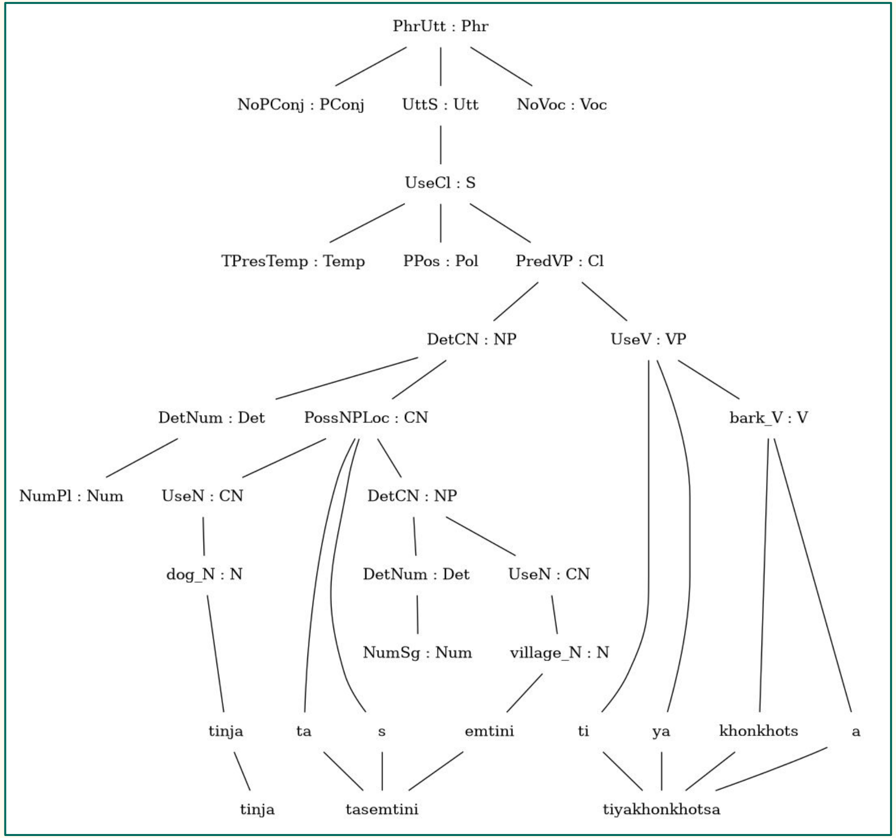
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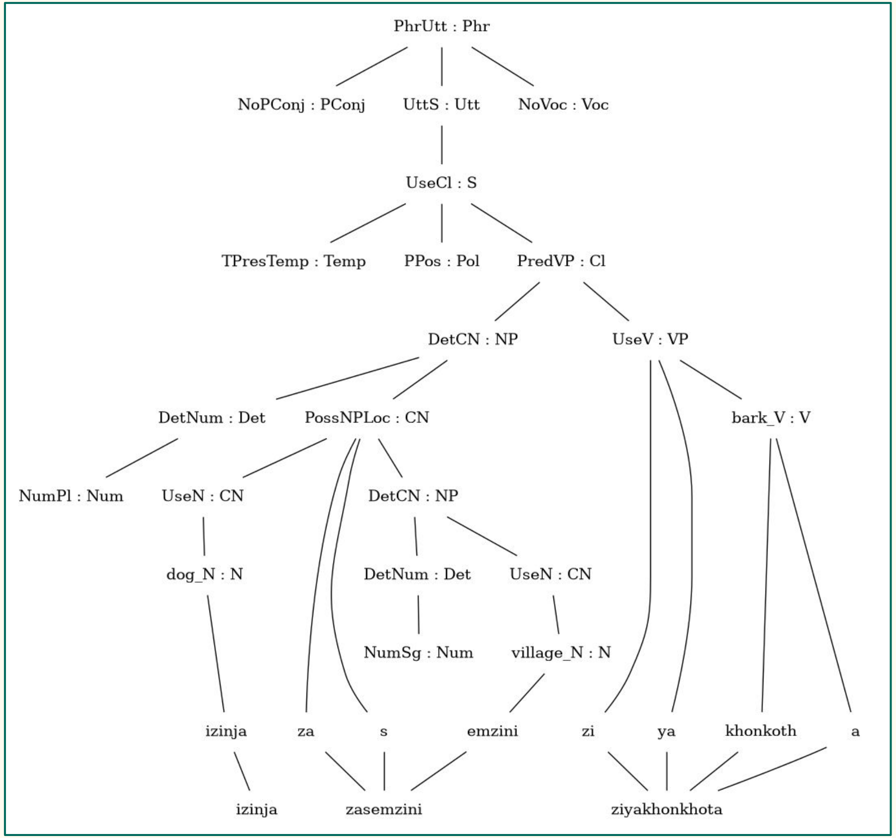
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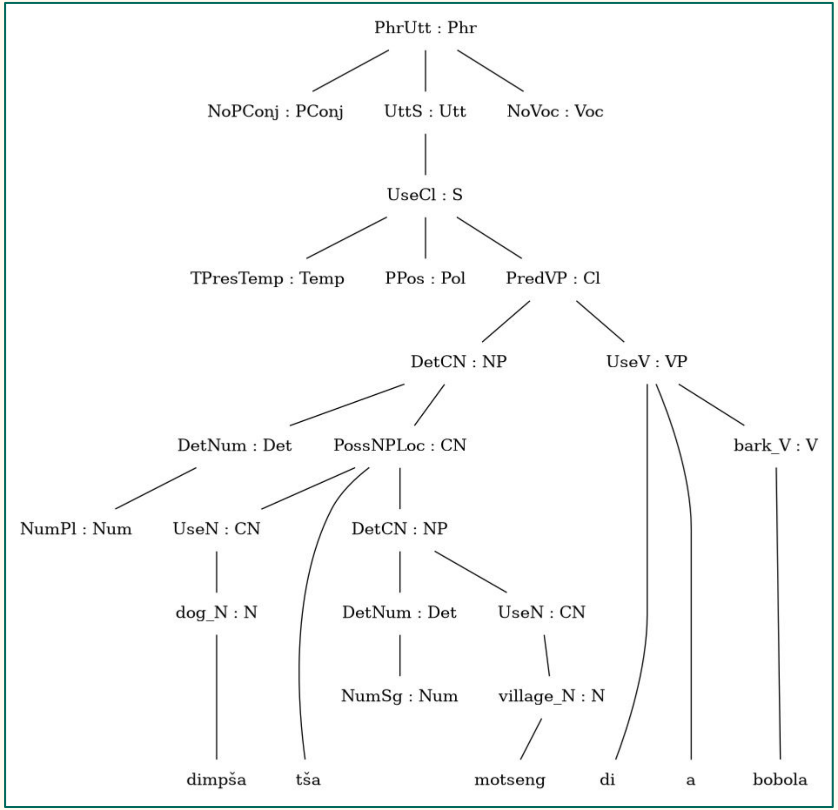
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bark_V : V
NumPl : Num
UseN : CN
DetCN : NP
dog_N : N
DetNum : Det
UseN : CN
NumSg : Num
village_N : N
dimpša
tša
motseng
di
a
bobola

Differences: Orthography and morphophonology

Northern Sotho

```
possConc : ClassGender -> Number -> Str
= \cg, num -> case cg of {
    C1_2 => case num of { Sg => "wa" ;
                          Pl => "ba" } ;
    -- ...
    C5_6 => case num of { Sg => "la" ;
                          Pl => "a" } ;
    -- ...
    C17 => case num of { _ => "ga" } ;
    C18 => case num of { _ => "ga" }
} ;
```

```
param
    NPForm = Absolute | Possessive | Locative ;
```

`Absolute = "Sezulu" ;`

"moithuti wa Sezulu"

Zulu

```
poss_concord : ClassGender => Number => RInit => Str =
  table {
    C1_2 => table {
      Sg => table {(RA|RC) => "wa" ; (RE|RI) => "we" ; (RO|RU) => "wo" } ;
      Pl => table {(RA|RC) => "ba" ; (RE|RI) => "be" ; (RO|RU) => "bo" }
    } ;
    -- ...
    C5_6 => table {
      Sg => table {(RA|RC) => "la" ; (RE|RI) => "le" ; (RO|RU) => "lo" } ;
      Pl => table {(RA|RC) => "a" ; (RE|RI) => "e" ; (RO|RU) => "o" }
    } ;
    -- ...
    C17 => table {
      _ => table {(RA|RC) => "kwa" ; (RE|RI) => "kwe" ; (RO|RU) => "ko" }
    }
  } ;
```

```
param
  NForm = NFull | NReduced | NPoss | NLoc ;
```

`NFull = "isiZulu" ; NReduced = "siZulu" ;`

"umfundi wesiZulu"

What next?

Document our process in such a way that it is possible to systematically

- identify the commonalities/similarities (structural and lexical);
- identify the differences (structural and lexical) between the new language and any or all the languages for which RGs already exist;
- develop a new RG in a modular, principled way;
- compile a development lexicon for use across the Bantu languages.

Continue to explore RG development (e.g. Khoekhoegowab) and application opportunities.